

UI Design Checklist

Complete UI Audit & Design Checklist

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Hasaballa AI Platform — Complete UI Audit & Design Checklist

Version 1.0 · 10 July 2026 · For the Design Team (owner: Deepanshi)

Covers every module in the platform that has a user-facing surface. For each: screens, components, user states, client requirements, whether a reference exists, and UX guidance.

0. Global / Cross-Cutting Requirements (apply to EVERY screen)

These are non-negotiable and must be baked into the design system before any individual screen is designed.

Requirement	Detail
Arabic-native, RTL-first	The primary language is Arabic. Every layout is right-to-left by default — not an LTR layout mirrored as an afterthought. Navigation, icons, progress bars, sliders and timelines all flow right-to-left.
Arabic typography	Must render diacritics, hamza, and mixed Arabic/Latin/numeral content correctly. Test every screen with real Arabic strings, not Lorem Ipsum.
Offline-first	No screen may imply or require an internet connection, except the explicit Publish flow. No spinners waiting on a network.
Desktop application	PySide6/Qt on Windows. Design for a resizable desktop window (min ~1280×720), not mobile.
Long-running operations	Generation can take 45+ minutes. Every generation screen needs progress, time-remaining, and cancel. Never a frozen window.
Cultural/religious compliance	Visual tone must respect the client's values. Compliance-related toggles must be visible and clearly labelled, never buried.

Standard user states — design ALL of these for every screen

Empty · Loading · Processing (with % + ETA + cancel) · Success · Error (with recovery action) · Disabled · Partial/Warning · Offline

The single most common gap in AI-product UI is the Processing and Error states. Because our pipeline is slow and multi-stage, these are the states users will spend the most time looking at. Design them first, not last.

1. Hasaballa GPT — Chat / Main Window

Status: Built (Done) · Reference:  Client provided (Platform doc) · Needs: Design polish only

Item	Detail
Screens	Main chat window; Saved Projects panel
Components	15,000-char script input box · Send button · Mic button (STT) · Reference Image insert · Reference Audio insert · Aspect-ratio selector (16:9 / 9:16 / 1:1) · Saved Projects list · Message/response display

States	Empty (no script) · Typing · Recording (mic active, waveform) · Character-count warning near 15k · Processing · Error (mic unavailable)
Client req.	15,000-character limit; mic input; reference image + audio attachment; dimension selector
UX notes	Character counter must be visible before the limit is hit. Mic recording needs an unmistakable active/recording indicator. Attachment thumbnails should be removable.

2. Timeline Editor (the largest UI surface)

Status: Core built, in Testing/Validation · **Reference:** ⚠️ Partial — standard video-editor conventions assumed; **no client mockup** · **Needs:** Design from scratch, benchmarked against known editors

Item	Detail
Screens	Timeline editor main view · Clip properties panel · Transition picker · Export dialog
Components	Canvas + time ruler · Multi-track layout (video / audio / subtitle) · Clip items (colour-coded) · Playhead scrubber · Drag-drop · Trim handles · Split (right-click) · Zoom control (1x–100x) · Snap-to-grid · Multi-clip selection · Frame thumbnails on clips · Undo/Redo · Transition objects (dissolve/fade/cross-fade/cut) · Per-clip volume envelope + fade handles · Ken Burns pan/zoom controls · Picture-in-Picture track · Chroma-key controls
States	Empty timeline · Loading project · Clip selected · Multi-select · Dragging · Trimming · Rendering (progress) · Autosave indicator · Recovery prompt (after crash) · Missing-media/placeholder clip
Client req.	Multi-track editing; undo/redo; autosave every 30s + on tool transition, last 50 autosaves retained; session-resume prompt on reopen; placeholder clips for skipped scenes (user can insert own video); aspect-ratio support; independent subtitle layer; PiP
UX notes	RTL timeline is the single biggest design question — does the playhead move right-to-left? Recommendation: keep the timeline itself LTR (universal editing convention, matches video time), but make all surrounding panels/labels/menus RTL. This must be confirmed with the client explicitly. Autosave needs a quiet, non-intrusive indicator. Undo/redo must be reachable by keyboard.

3. Image Generation

Status: In progress (Charvina) · **Reference:** ❌ None — **design from scratch** · **Needs:** Full design

Item	Detail
Screens	Image generation panel · Per-scene image review/gallery · Regeneration dialog
Components	Prompt display (Arabic) · Seed field (+ lock for reproducibility) · Batch queue (14 scenes) · Per-scene image grid · Regenerate button (per image) · Approve/reject per image · Reference-image slot
States	Queued · Generating (per-scene progress, N of 14) · Success · Failed (with retry) · Rejected by compliance filter (auto-regenerating) · Manual review needed
Client req.	Offline generation; seed-based reproducibility; 14-scene batch; image compliance — no nudity/immodest imagery, auto-reject + regenerate
UX notes	Show the batch as a grid with per-scene status , not a single spinner — users need to see which scene failed. Compliance rejections

	should be handled <i>silently and automatically</i> (regenerate in modest form); only surface to the user if it fails repeatedly. Never display a rejected non-compliant image to the user, even briefly.
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4. Character Pack Manager

Status: Data model Done; UI not built · **Reference:**  Client spec (8 ref images/char, weights, voice pairing) · **Needs:** Design from scratch

Item	Detail
Screens	Character list · Character detail/editor · Import/export dialog
Components	Character card grid · 8 reference-image slots per character · Per-image weight slider · Image↔voice pairing control · Add/remove character · JSON import/export
States	Empty (no characters) · Incomplete (fewer than 8 images) · Complete · Identity-conflict warning
Client req.	8 reference images per character; per-image weighting; image–voice pair link; JSON export/import
UX notes	Make the 8 image slots visually explicit (8 placeholders), so users understand the expectation. Weight sliders need a sensible default so users can ignore them.

5. Voice / TTS & Voice Cloning

Status: In progress (Diya) · **Reference:**  Partial (dialect + speed specs given) · **Needs:** Design from scratch

Item	Detail
Screens	Voice selection panel · Voice speed control · Voice preview
Components	Voice library browser (12 named voice types) · Dialect selector (21 dialects) · Per-character voice assignment · Speed slider 0.5x–2.0x + preset buttons (slow / normal / fast / sprint) · Preview play button per voice · Reference-audio upload (cloning) · Waveform preview
States	No voice selected · Previewing (playing) · Generating · Voice-clone processing · Preview sample missing (library still being curated) · Dialect unsupported/low-quality warning
Client req.	12 named voice types mapped to dialect clusters, with preview samples; per-character speed 0.5x–2.0x with presets; inline <code>[voice.speed=X]</code> command support; 21 Arabic dialects
UX notes	Not all 21 dialects have equal quality — the UI must communicate dialect quality honestly (e.g. a quality indicator per dialect) rather than presenting all 21 as equivalent. Preview samples are still being collected; design an "audio preview unavailable yet" state.

6. Lip Sync

Status: In progress (Dhruvi) · **Reference:**  Modes specified, no mockup · **Needs:** Light UI

Item	Detail
Screens	Lip-sync mode selector (within timeline clip properties)
Components	4 rendering-mode selector (natural / radio / phone / megaphone) · Re-render trigger · Sync preview · isTalking indicator per character

States	Not applied · Processing (slow — needs progress) · Applied · Re-rendering (mode changed) · Failed / drift detected
Client req.	4 voice modes mapped to LatentSync presets; auto-activation via <code>isTalking</code> flag; re-render on mode change
UX notes	Changing mode triggers an expensive re-render — warn the user before re-rendering , don't do it silently on every dropdown change.

7. Audio: Layering, Sound Library & Compliance

Status: In progress (Roshni) · **Reference:**  Strong client spec (16 categories, compliance rules) · **Needs:** Design from scratch

Item	Detail
Screens	Sound library browser · Audio layering / mixer panel · Religious Compliance settings · Demucs standalone audio tool
Components	Sound library browser: 16 FX categories (Emotional, Military, Tension, Ambience, City, Nature, Crowd, Broadcast, Chase, Horses/Combat, Religious, Arab Cultural, Era Ambiences, Foley...) with era + region tags · Search/filter · Preview play · Religious Compliance toggle (music / oud / tarab / neutral per-asset flags) · Per-track mute/solo · Volume sliders · Fade in/out handles · Waveform preview · Auto-duck indicator (−18 dB under dialogue) · LUFS meter (−14 target)
States	Library loading · Category empty (still being curated — real state today) · Preview playing · Asset blocked by compliance (visibly indicated, with reason) · Mixing/processing
Client req.	Critical: compliance toggle must automatically block non-compliant sounds (e.g. no background music during Adhan / Quran recitation / Dua); per-asset flags music/oud/tarab/neutral; 16 categories × ≥50 samples; era + region tagging
UX notes	The compliance toggle is a flagship client requirement — give it prominence and make its effect legible ("12 assets hidden by compliance settings"). Blocked assets should be clearly explained, not silently missing. Several categories are still incomplete — design an honest "category not yet populated" empty state.

8. Smart Director (Pipeline Orchestrator)

Status: In progress (Roshni) · **Reference:**  Partial (Auto/Manual modes specified) · **Needs:** Design from scratch — high importance

Item	Detail
Screens	Smart Director main panel · Visual progress tracker · Manual step-through view
Components	Auto vs Manual mode switch · Step-by-step progress panel (current step highlighted, completed ✓, skipped greyed) · Per-stage skip controls (image / animation / voice / compilation) · Scene queue (14 scenes) · Per-scene status · Overall ETA · Cancel / pause
States	Idle · Running (per-stage + per-scene progress) · Paused · Stage skipped · Stage failed (retry / skip / abort) · Complete
Client req.	Auto + Manual modes; [[skip]] flag honoured per stage; step-by-step visual progress for Manual mode
UX notes	This is the screen users will stare at for 45 minutes — it deserves the most design attention of any screen in the product. Show <i>both</i> overall progress and current-scene detail. A failure at scene 12 of 14

	must not discard the first 11. Make "what is it doing right now" unmistakable.
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9. Video / Motion Generation (WAN 2.2) & BIE

Status: WAN in progress (Rudra); BIE starting (Rudra) · **Reference:** ❌ None · **Needs:** Minimal UI (mostly backend)

Item	Detail
Screens	Motion/animation settings (within scene properties) · (BIE runs silently in backend)
Components	Camera-effect selector (pan / zoom / dolly / rack focus) · Body-motion toggle · Era-based motion style (auto, silent) · Cinematic FX layer (smoke/haze/explosions) · Quality-guard status
States	Not animated · Generating (slow, GPU) · Complete · Quality guard: drift/artefact detected → auto re-generating affected frames · GPU busy/queued
Client req.	Era-appropriate motion applied automatically and silently; auto-quality guard re-generates bad frames with no user action
UX notes	BIE and era-motion are explicitly "runs silently in backend" — do not expose them as user controls. The only UI need is a status indicator. Because these need the shared GPU, add a "waiting for GPU" queued state.

10. Subtitles, Captions & Translation

Status: To do (Deepanshi / Diya) · **Reference:** ✅ Spec'd · **Needs:** Design from scratch

Item	Detail
Screens	Subtitle editor · Caption styling panel · Language/dubbing selector
Components	Manual subtitle block editor · Auto RTL Arabic alignment · Font / size / colour picker · Timing shift (±X ms) · SRT import/export · Burn-in toggle at export · Auto-sync with audio · Dual-language caption layer (Arabic + English + 3 others) · Per-scene language override
States	No subtitles · Auto-generating (from Whisper) · Editing · Out-of-sync warning · Translating (NLLB) · Translation failed
Client req.	Independent subtitle layer with styling; RTL alignment; SRT import/export; burn-in toggle; multilingual captions via NLLB-200
UX notes	Arabic subtitle styling preview must render in actual Arabic with diacritics — a Latin preview is useless here. Timing-shift needs live preview.

11. Export & Render

Status: To do (Deepanshi / Diya) · **Reference:** ✅ Spec'd (formats, ratios, presets) · **Needs:** Design from scratch

Item	Detail
Screens	Export dialog · Render queue · Render preview
Components	Multi-ratio export (16:9 + 9:16 + 1:1 simultaneously, all 4K) · Format selector (MP4 / MOV / WebM) · Bitrate presets (YouTube 4K / Instagram / WhatsApp) · Low-res proxy preview (¼ res) · Render queue with per-item progress · Audio mix-down to stereo master · Subtitle burn-in toggle

States	Idle · Queued · Rendering (per-ratio progress) · Paused · Complete (with "open folder") · Failed (per-ratio — one can fail while others succeed) · Disk-space warning
Client req.	Simultaneous 4K export in all three ratios, each maintaining 4K independently; render queue + progress tracking; platform bitrate presets
UX notes	Three simultaneous 4K renders is heavy — show per-ratio progress separately . Warn on low disk space <i>before</i> starting. "Open output folder" on success is a small touch that users always want.

12. Project Management (Save / Load / Backup)

Status: In progress (Diya) · Reference:  Spec'd · Needs: Standard dialogs

Item	Detail
Screens	New / Open / Save dialogs · Saved Projects panel · Backup & restore browser
Components	Project list (with thumbnails + dates) · New/Open/Save/Save-As · Folder structure (project.json + /assets + /exports) · Backup → timestamped .zip · Restore from archive · Backup browser · Autosave-recovery prompt
States	No projects (first run) · Loading · Unsaved changes (warn on close) · Autosave recovery available · Backup in progress · Restore confirmation · Corrupt project (graceful error)
Client req.	Crash-safe atomic writes; last 50 autosaves retained; session-resume prompt on reopen; full project backup to zip + restore
UX notes	The autosave-recovery prompt on reopen is a client-specified requirement — design it explicitly. Restore is destructive: require confirmation.

13. Import & Media

Status: To do (Deepanshi) · Reference:  Spec'd · Needs: Light design

Item	Detail
Screens	Import dialog (or drag-drop onto timeline)
Components	Drag-and-drop zone · File browser · Supported: MP4 / MOV / PNG / JPG / WAV / MP3 · ffprobe codec detection · Media bin
States	Empty · Dragging-over (highlight) · Importing (progress) · Unsupported codec (clear error) · Import failed
UX notes	Codec errors must say <i>what</i> is wrong and <i>what to do</i> , not just "failed".

14. Settings, Compliance & Offline Controls

Status: Mixed · Reference:  Client-specified · Needs: Design from scratch

Item	Detail
Screens	Settings · Smart Internet Access panel · Compliance settings · Model/paths config
Components	Offline-first internet toggle (never auto-connects) · "Disconnect" / "Return to Offline" buttons · Religious compliance settings · Image

	compliance / modesty filter settings (sensitivity threshold) · Model path configuration · Language/RTL settings
States	Offline (default) · Online (explicit, clearly indicated) · Connecting · Compliance filter active/inactive · Model missing (clear guidance)
Client req.	Offline by default; the app must NEVER auto-connect to the internet ; explicit Disconnect / Return-to-Offline controls; religious + image compliance safeguards
UX notes	The online/offline state must be globally visible at all times (e.g. a persistent status indicator), because "am I offline?" is the client's core trust question. Going online should require deliberate action and be visually loud.

15. Publishing (the ONLY online surface)

Status: To do (Deepanshi) · **Reference:**  Spec'd · **Needs:** Design from scratch

Item	Detail
Screens	YouTube upload · OAuth consent flow · AdSense/Analytics dashboard
Components	OAuth 2.0 login flow · Arabic title / description / tags fields · Upload progress · Retry on network failure · Revenue + CPM dashboard
States	Not authenticated · Authenticating · Uploading (progress) · Upload failed (retry) · Published (with link) · Token expired · Offline (upload unavailable — expected default state)
Client req.	OAuth 2.0 (YouTube Data API v3); Arabic metadata; upload progress; retry on failure; secure token storage; AdSense revenue + CPM display
UX notes	This is the only feature that requires internet — make that explicit and consensual. Never store tokens in plain text. Since the app is offline by default, the upload screen's default state is "offline / unavailable" — design that first.

16. Standalone Tools

Status: Demucs Done; Whisper in validation · **Reference:**  Partial · **Needs:** Light design

Item	Detail
Screens	Demucs audio tool · Whisper STT/SRT tool
Components	Demucs: raw audio in → noise removal → vocal separation → EQ → -14 LUFS out; waveform; before/after preview. Whisper: audio in → transcript + SRT out; language/dialect selector; accuracy display
States	Empty · Processing · Complete (download/save) · Failed
UX notes	These double as standalone tools <i>and</i> pipeline stages — the standalone UI should be simple and self-contained.

Verification Pass

Every module in the roadmap checked for a UI surface:

Module	UI?	Covered in
Timeline (canvas, clips, interactions, transitions, autosave, filters, PiP, chroma, Ken Burns, snapping, smart edit)	Yes	§2
Hasaballa GPT Core / Chat Window	Yes	§1

Image Generation (SDXL/FLUX)	Yes	§3
InstantID / PuLID / ControlNet identity	No direct UI (backend) — surfaces as image quality/regeneration	§3, §9
Character Pack Data Model	Yes	§4
TTS / Voice Cloning, Voice Library, Voice Speed (7-Q)	Yes	§5
Inline voice-speed command parser	No UI (text command) — but speed UI must stay in sync	§5
Lip Sync (activation, 4 modes, timeline)	Yes	§6
Audio Layering, Sound Library (16 cats), Religious Compliance, Dialogue FX (7-P), Voice QA (7-O)	Yes	§7
Smart Director (Auto/Manual, progress tracker, auto-skip)	Yes	§8
WAN 2.2 / Body animation / Camera FX / Era motion / Quality guard / BIE	Minimal (silent backend + status)	§9
Subtitles, Captions, NLLB Translation, Dubbing	Yes	§10
Export, Multi-ratio, Render preview, Format selector	Yes	§11
Project Save/Load, Backup & Restore	Yes	§12
External Media Import	Yes	§13
Smart Internet Toggle, Settings, Image Compliance filter	Yes	§14
YouTube Upload / OAuth / AdSense	Yes	§15
Demucs standalone, Whisper STT	Yes	§16
Scenario Parser, Qwen LLM, Real-ESRGAN, Whisper, Arabic prompts, Video assembler, Model router, Parallel execution, Depth parallax, VRAM optimisation	No UI — pure backend	— (correctly excluded)
Image enhancement / Image animation		

Cross-cutting confirmed: RTL/Arabic  · all 8 user states  · offline-first indicator  · long-operation progress/cancel  · compliance (audio + image)  · desktop/Qt constraints 

Open questions for the client — must be answered before design is finalised

1. **Timeline direction:** should the timeline track itself be RTL, or stay LTR (standard editing convention) with RTL surrounding UI? (*Recommendation: LTR timeline, RTL chrome.*)
2. **Reference mockups:** the client has provided the platform requirements doc but **no visual designs**. Confirm whether they will supply branding/visual direction, or whether we design from scratch and present for approval.
3. **Dialect quality display:** how transparent should the UI be about which of the 21 dialects have lower TTS quality?

Addendum — Gaps Found in Final Verification (NOT previously documented)

These surfaced on the second pass. They are real requirements implied by the project but **absent from both the roadmap and the original audit**. They need owners and dates.

#	Missing item	Why it matters	Severity
A1	First-run / setup wizard	The app needs multi-GB model weights placed correctly, a CUDA-capable GPU, and RDP config. There is currently no onboarding screen — a user's first launch would simply fail with a stack trace. The roadmap has a "model placement guide"	 High — this is the client's <i>literal first experience</i> of the product

		(a <i>document</i>), but no UI for it. Needs: model-path picker, "models missing" detection, GPU/VRAM check, guided first-run.	
A2	Completion notification	Renders take 45+ minutes. Users will alt-tab away. Needs a desktop notification / sound / taskbar flash on completion or failure.	● Medium
A3	Prevent system sleep during render	Windows will sleep mid-render and destroy a 45-minute job. Needs a keep-awake lock plus a UI indicator.	● Medium
A4	Image Compliance review queue	If auto-regeneration fails repeatedly, an image needs human review. §3 covers the automatic path; there is no UI for the manual fallback . Must never display the non-compliant image itself — show the scene, the reason, and offer "regenerate / edit prompt / skip".	● Medium
A5	Arabic ⇌ English UI language toggle	Arabic is primary, but developers/QA (and possibly some client staff) need English. Not currently specified either way. Confirm with client.	● Confirm
A6	Keyboard shortcuts	A timeline editor without shortcuts (space=play, ctrl+Z, S=split, del) is unusable professionally. Not documented anywhere.	● Medium
A7	"GPU busy" global state	One shared RTX 5090; multiple stages queue for it. Users need to see " <i>waiting for GPU — 2 jobs ahead</i> " rather than an apparently frozen app.	● Medium
A8	Error-log export / "report a problem"	For an offline app there's no telemetry. When something breaks on the client's machine, we need him to send us something. One-click log export.	● Low
A9	Storage management	Projects with 4K intermediates will consume hundreds of GB. Needs disk-usage display and a cache-clear control.	● Low
A10	Estimated time remaining (ETA)	Specified loosely as "progress" but never as ETA. For a 45-minute operation, "72% complete" without a time estimate is close to useless.	● Medium

Developer-facing implementation notes (from the requirements, worth flagging to the design team)

- **Arabic rendering is not a font choice — it's a pipeline.** Every Arabic string must pass through `arabic_reshaper` + `python-bidi` before display, or it renders as disconnected, backwards letters. Design mockups must use *real* reshaped Arabic, not placeholder text.
- **Long operations must never block the Qt main thread** — everything heavy runs on a worker thread, or the entire UI freezes and Windows marks it "Not Responding".

- **The offline/online indicator is a trust feature, not a status icon.** The client's core anxiety is data leaving his machine. It should be persistent and unmissable.
- **Autosave/recovery is client-specified in detail** (30s interval, last 50 autosaves, resume prompt) — treat it as a first-class feature, not an afterthought.

Design priority order (aligned to the delivery plan)

4. **Smart Director progress tracker** (§8) — highest user-time-on-screen
5. **Timeline editor** (§2) — largest surface, already partly built
6. **Image generation + compliance** (§3) — client's active concern
7. **Audio/sound library + religious compliance** (§7) — flagship client requirement
8. Voice/TTS controls (§5) · Export (§11) · Project management (§12)
9. Settings/offline (§14) · Subtitles (§10) · Import (§13)
10. Publishing (§15) · Standalone tools (§16)